# Jacomo Corrieri

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## EDUCATION

**University of Florida**, *Gainesville, FL* *Expected May 2027*
Master of Science (M.S.) in Computer Science
**GPA: 3.83/4.0**

**University of North Florida**, *Jacksonville, FL* *May 2025*
Bachelor of Science (B.S.) in Computer Science, Minors in Mathematics and International Business
**GPA: 3.99/4.0** | *Summa Cum Laude* | Hicks Honors College

## WORK EXPERIENCE

**Florida Blue**, *Jacksonville, FL* *May 2026 – Present*
Information Technology Intern – Software Engineering
• Develop an application to manage, standardize, and track relational migration resources using React.js/Express.js and Java Spring Boot, targeting **8–10 hours** of weekly time savings by eliminating manual database operations.
• Validate backend service components with SQLite-based unit tests, benchmark SQL Server query response times, and test deployments using Jenkins and OpenShift while maintaining system design documentation and Agile user stories.
• Present project architecture design and functionality to cross-functional team leaders, including senior leadership.

**UFIT Research Computing**, *Gainesville, FL* *Jan 2026 – May 2026*
Research Computing Support Intern
• Built a Python CLI and Django REST API to query **800M+ HiPerGator usage records**; containerized the API and PostgreSQL database with Podman/Docker and secured endpoints using RS256 JWT authentication.
• Accelerated HPC module deprecation checks by **10–30×** by caching read-heavy filesystem module data in PostgreSQL tables, refreshed hourly via Slurm cron jobs, and automated daily usage data ingestion for **3700+ users**.

## RESEARCH

**ICTAI 2026** - *Scalable Multi-robot Coverage with Connectivity Maintenance Via Deep Reinforcement Learning*
Corrieri Jacomo, Kulbaka Iliya, Ghosh Anirban, Dutta Ayan
• Simulated up to 50 autonomous robots with Python and Gymnasium, achieving **5x more scalability** than prior work.
• Developed a decentralized CNN model with PyTorch, resulting in **15% greater area coverage** compared to our baseline.
• Visualized performance metrics with Matplotlib and TensorBoard for data analysis and model evaluations.

## PROJECTS

**Tiny DNS Cache** *Sep 2026 – Present*
• Build a multithreaded C++20 DNS cache using Unix sockets, thread pooling, and a bounded producer-consumer queue.
• Design a thread-safe templated cache with a shared mutex, plus a RAII SQLite wrapper and CTest (CMake) tests.
• Develop record lookup with recursive CTEs for CNAME/NS chains, zero-copy views, and binary IPv4/IPv6 decoding.

**MoTorch Neural Network Library** *Apr 2026 – Jul 2026*
• Created a PyTorch-inspired deep learning framework with neural network modules, optimizers, and loss functions.
• Implemented a reverse-mode autodiff engine for tensors with a dynamic computational graph for backpropagation.
• Automated CI and testing with GitHub Actions and PyTest, benchmarking models with up to **32.6 million parameters**.

**UF Student Marketplace** *Feb 2026 – Jul 2026*
• Engineered a full-stack Angular/TypeScript and Go application using WebSockets for real-time messaging.
• Replaced database blobs with scalable cloud-based object storage by integrating AWS S3 with a Gin REST API.
• Designed and indexed relational schemas with GORM and SQLite for listings, messaging, and user workflows.

## SKILLS

**Languages:** Python, C++, Go, Java, TypeScript, SQL
**Libraries:** FastAPI, Django, Express, React, CUDA, PyTorch, NumPy, Gymnasium (Pygame)
**Infra & Tools:** Linux, Slurm, Git, GitHub Actions, Podman, PostgreSQL, SQLite, AWS S3, Codex, CMake, Jenkins, OpenShift